International Journal of Forecasting 42 (2026) 616–639

Contents lists available at ScienceDirect

International Journal of Forecasting

journal homepage: www.elsevier.com/locate/ijforecast

Leveraging image-based generative adversarial networks for time series generation

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ARTICLE INFO

Keywords:
Extended Intertemporal Return Plots
Generative Adversarial Networks
Time Series Imaging
Data augmentation
Stochastic Inversion
Diffusion

ABSTRACT

Generative models for images have gained significant attention in computer vision and natural language processing, due to their ability to generate realistic samples from complex data distributions. To leverage the advances of image-based generative models for the time series domain, we propose a two-dimensional image representation for time series, called the extended intertemporal return plot (XIRP). Our approach captures the intertemporal time series dynamics in a scale-invariant and invertible way, reducing training time and improving sample quality. We benchmark synthetic XIRPs obtained by an off-the-shelf Wasserstein generative adversarial network with gradient penalty against other image representations and models regarding sample similarity, predictive ability, similarity of moments, and forecast enhancement metrics. Our novel, validated image representation for time series consistently and significantly outperforms the state-of-the-art recurrent neural network and diffusion-based generative model in terms of predictive ability. Synthesis results obtained by alternative image-based representations additionally exceed the benchmarks regarding similarity across moments. Further, we introduce an improved stochastic inversion to substantially improve simulation quality regardless of the representation and provide the prospect of transfer potential in other domains.

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1. Introduction

Time series simulation is fundamental to financial, economic, and engineering applications and serves the purpose of evaluating products or making probabilistic predictions. Financial institutions run complex Monte Carlo simulations to quantify risks, price financial products and commodities, or identify patterns for trading strategies (Drew, 2012). The models often require large amounts of data to make accurate predictions, and real-world data may be scarce. Simulating time series data enables the creation of synthetic data that resemble real-world data for various purposes, including modeling, prediction, and analysis. In financial applications, synthetic time series facilitate the modeling of stock prices, interest rates, and other financial instruments to make informed predictions about future market trends and evaluate the performance of investment strategies. In economic applications, synthetic time series data can support estimating variables such as GDP, inflation, and unemployment to strengthen economic predictions and evaluate the impact of shocks or policy changes. Also, in engineering applications, the simulation of time series data is used to model the performance of power grids, transportation networks, and communication systems. High-quality simulated data enables engineers to assess system performance and monitor the impact of design changes.

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The numerical results gathered in this paper could not be reproduced owing to the computational requirements involved.

https://doi.org/10.1016/j.ijforecast.2025.09.004
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Simulation techniques often rely on parametric approaches to generate time series. In the first step, parametric models assume a reasonably similar data-generating process. Afterwards, the models estimate coefficients for a forecasting model that has been selected based on the assumption of the process. However, fitting a forecasting model across time is risky, especially since time series can visit different regimes with opposing characteristics. Despite the drawback that the number of regimes is often specified beforehand and excludes the possibility of other unobserved regimes in unseen data, financial applications frequently use regime-switching models (Herrera et al., 2018; Scharth & Medeiros, 2009). Once multiple time series are involved, training and calibrating the model becomes tedious, since models rely on estimating correlations between many features, and regime switches result in non-stationary correlations over time (Schäfer & Guhr, 2010). However, the higher the expectation of the model's accuracy and generalization, the more training data required. Further, some behavior is rare and may only occur scarcely but is highly important in the business context. A miscalibrated model due to a data shortage might have far-reaching consequences. Solutions to the problems include a shallower network, sacrificing some forecast accuracy, or synthesizing additional data that can help to fit parametric models.

Artificial data that resemble real-world data are frequently used in computer vision to train models and algorithms (Calimeri et al., 2017; Krizhevsky et al., 2012; Wang et al., 2017). Synthetic data have several advantages over real-world data, such as being cost-effective, easier to obtain, and often free from privacy and ethical issues. Further, synthetic data allow for balancing datasets, creating datasets with specific characteristics, and training models in cases where real-world data are scarce or unavailable. Recently, deep learning models augmented the set of models to generate new images that are similar in appearance to a given training set of images. The dominant classes of the models do not assume that the generated images come from a particular distribution, such as a Gaussian or a uniform distribution. Instead, they learn the underlying distribution of the training data via adversarial training of a generative and a discriminative network. They generate diverse and complex images similar to the training data and learn complex and multi-modal distributions.

In particular, Karras et al. (2019) demonstrated high-quality synthetic data of faces with disentangled, controllable features. While generative adversarial networks (GANs) (Goodfellow et al., 2014) have long been a quasi-standard for generative models, the new class of diffusion models has recently surpassed GANs at generating artificial images in a benchmark study (Dhariwal & Nichol, 2021a). The current successes of text-to-image models impressively prove the capabilities of conditional image generation (Ramesh et al., 2021) and can contribute to art, design, and content creation. The image-based generative GAN compensates for the drawbacks of classic parametric models by not assuming the data-generating process, allowing it to incorporate non-stationarity easily and making no assumptions about regimes or intertemporal relationships. A crucial step, however, is to find an efficient image representation for the time series to leverage existing image-based approaches and benefit from their modeling advantages and generative abilities. Therefore, it needs to represent the time series in a two-dimensional space that preserves the characteristics of the time series and allows for direct inversion after the synthesis process.

Here, we (1) propose a novel methodology for univariate time series, called the extended intertemporal return plot (XIRP), capable of capturing the dynamics of time series; (2) validate our approach with simulated and real data; (3) perform a benchmark test with two state-of-the-art generative models; (4) provide guidance for a stochastic inversion of the image representation to increase the predictive ability of the synthesis for real data; and (5) show that an effective image-based augmentation enhances forecasting results.

The new image representation considers scale-invariant intertemporal relationships stored in the off-diagonal of the image, while the diagonal contains the initial time series. So, it preserves both intertemporal relationships and raw time series values. This results in three significant advantages for the proposed approach:

Firstly, the XIRP captures the intertemporal dynamics of a time series better than other representations. Motivated by principles from finance, it encodes the initial time series and its intertemporal returns in a single image, allowing for easy deterministic and stochastic inversion.

Secondly, the proposed XIRP proves competitive and surpasses other image representations regarding predictive ability and similarity in moments across a range of synthetic and real datasets.

Thirdly, our approach allows for a large reduction in model complexity. Instead of relying on multiple steps to encode and decode, the model processes the XIRP directly and samples new image representations, which are then inverted to obtain a time series.

Our results have multiple implications for research and practice. On the research side, our approach motivates the use of image-based representations for various advanced generative models. The XIRP is suitable for all types of image-based generative models, and its design extends the field of application beyond generation toward regression and classification tasks. The XIRP enables the exploration, review, and improvement of engineering, medical, and energy applications that have so far relied on other image representations, e.g. for fault diagnosis (Cui et al., 2022), biosignal classification (de Paula et al., 2023), and simulations of scenarios for renewable energy generation (Wang et al., 2022). Augmented data based on XIRPs can potentially improve any parametric and nonparametric model in terms of the generalization ability and overall performance. The results can be directly transferred to make forecasting models more robust against adversarial attacks (Sarkar et al., 2020) and to improve forecasting accuracy (Elalem et al., 2022; Zhang et al., 2022).

To our knowledge, this paper is the first to evaluate an invertible image representation of time series for generative models in terms of similarity and predictive ability, comparing the results to state-of-the-art benchmarks.

We hope that our contribution intensifies the research with newly developed image-based generative models and encourages practitioners to incorporate data augmentation for time series as a standard part of the model development workflow.

2. Background

This section provides a comprehensive introduction to generative adversarial networks and time series images, which are the foundation of the paper. It also introduces concepts relevant to our research questions and hypotheses and aims to provide a thorough understanding of the context. We hope that the discussion enables readers to appreciate the impact and contribution of our propositions in the areas of time series representations, generative models, and data augmentation.

2.1. Generative adversarial networks

From the generative aspect, traditional statistical methods and machine learning techniques often fail to capture the temporal dependencies, non-linear patterns, and multi-model distributions of time series data (Lipton et al., 2017). GANs have proven effective at addressing these issues. They emerged as a powerful tool for learning the underlying data distribution and generating realistic synthetic samples. The generative concept relies on two neural networks, the generator G and discriminator D, trained in an adversarial setting. The generator takes a noise vector sampled from a probability distribution as input and generates synthetic samples. In contrast, the discriminator receives real and synthetic samples and attempts to distinguish between them. The generator and discriminator networks are trained iteratively in a minimax game until the generator can produce synthetic samples indistinguishable from real data. The objective function of the adversarial game between G and D is given by

$$\min_G \max_D \mathop{\mathbb{E}}_{x\sim\mathbb{P}_r}[log(D(x))] + \mathop{\mathbb{E}}_{\tilde{x}\sim\mathbb{P}_g}[log(1-D(\tilde{x}))] \quad (1)$$

where the discriminator and generator try to maximize and minimize (3), respectively. $D(x)$ is the estimate of the discriminator that x is real, while $\mathbb{P}_r$ denotes the real data distribution. The generated data are denoted by $\tilde{x}$, with $\tilde{x} = G(z)$ and $z \sim p(z)$, where p denotes some distribution such as uniform or Gaussian. $\mathbb{P}_g$ is the distribution of the generated data or, put differently, the model distribution. The adversarial setting does not require explicit modeling of the probability distribution of the training data, which can be difficult or impossible for high-dimensional and complex data. Instead, GANs implicitly learn the distribution by generating synthetic samples that match the real ones. According to Bengio (2009), the primary motivation for the design of GANs is that the learning process requires neither approximate inference nor approximation of a partition function gradient. If the objective function of the discriminator is convex, the procedure is guaranteed to converge and is asymptotically consistent. In practice, however, the above vanilla GAN suffers from stability problems and is difficult to train (Goodfellow et al., 2014), especially when the discriminator's objective is not convex.

Arjovsky et al. (2017) address this problem by proposing a Wasserstein GAN (WGAN), which shows more stable training behavior and correlation between the value of the objective and sample quality. An intuitive explanation for the Wasserstein distance as a loss function is its synonym, the earth mover's distance. It can be related to the minimum cost of transporting mass to transform distribution q into distribution p, where the cost is the mass times transport distance. The Wasserstein metric changes the objective function (1) to

$$\min_G \max_{D\epsilon\mathcal{D}} \mathop{\mathbb{E}}_{x\sim\mathbb{P}_r}[D(x)] - \mathop{\mathbb{E}}_{\tilde{x}\sim\mathbb{P}_g}[D(\tilde{x})] \quad (2)$$

where $\mathcal{D}$ is defined as a set of 1-Lipschitz functions, and $\hat{x}$ is a randomly sampled subset. Instead of classifying generated images as real or fake, the WGAN with gradient penalty (WGAN-GP) replaces the discriminator model with a critic that scores the realness or fakeness of an image. Gulrajani et al. (2017) further improve training in their WGAN-GP by incorporating gradient penalty terms for the discriminative network. The penalty further alters the objective function to

$$\min_G \max_{D\epsilon\mathcal{D}} \mathop{\mathbb{E}}_{x\sim\mathbb{P}_r}[D(x)] - \mathop{\mathbb{E}}_{\tilde{x}\sim\mathbb{P}_g}[D(\tilde{x})] + \lambda \mathop{\mathbb{E}}_{\tilde{x}\sim\mathbb{P}_{\hat{x}}}[(\|\nabla_{\hat{x}}D(\hat{x})\|_2 - 1)^2] \quad (3)$$

The gradient penalty term $[(\|\nabla_{\hat{x}}D(\hat{x})\|_2-1)^2]$ opposes vanishing and exploding gradients. The major advantage of WGAN-GP is its improved convergence. Previous models optimizing Eq. (2) suffer from mode collapse, a scenario in which the generative model samples with little to no diversity. Advanced methods adapt the model architecture, as done by Allahyani et al. (2023), or improve the loss functions, as in the WGAN-GP. We refer the reader to Gulrajani et al. (2017) for more detailed explanations.

GANs need large amounts of training data to learn the underlying distribution of the data accurately. Insufficient or unrepresentative data produce unrealistic and low-quality synthetic samples. Further, GANs can suffer from instability and sensitivity to hyperparameter settings, making them challenging to tune and optimize. At the same time, the choice of architecture, optimization method, and regularization technique can significantly affect their performance and stability.

Whereas WGAN-GP relies on image input, other models rely on time series as inputs (Esteban et al., 2017; Yoon et al., 2019). These rely on recurrent neural networks (RNNs) and have already been applied in economic and financial settings for identifying trading strategies and enhancing predictive tasks in the market environments (Carvajal-Patiño & Ramos-Pollán, 2022; Wu et al., 2022). The most prominent model is TimeGAN (Yoon et al., 2019), which outperforms many related models, e.g. the ones proposed by Esteban et al. (2017) and Donahue et al. (2019). TimeGAN relies on various networks and combines supervised and unsupervised training to account for temporal correlations. TimeGAN learns a time series embedding and optimizes multiple loss functions.

Its architecture comprises an encoder/decoder structure to facilitate the process and a supervising network to preserve intertemporal dynamics. Thus, TimeGAN requires five different loss functions to control the generator, the discriminator, intertemporal relationships, moments, and the embedding loss. Encoder/decoder networks are obsolete for the WGAN-GP, since the proposed image representation captures temporal dynamics and omits an embedding loss due to its numeric invertibility. Further, the TimeGAN architecture assigns each network a different task, and each component is trained separately before starting joint training. Despite the slow training, TimeGAN is considered a proper benchmark for any generative time series model, due to its superior simulation results (Dash et al., 2020; Naveed et al., 2022; Yoon et al., 2019).

2.2. Diffusion models

Diffusion models have recently emerged as a competitive benchmark for generative modeling in both the image and time series domains (Dhariwal & Nichol, 2021b; Ho et al., 2020a; Li et al., 2023). In contrast to TimeGAN, which involves multiple networks and sequential training phases, diffusion models rely on a conceptually simpler yet powerful framework based on iterative denoising. They define a two-stage generative process: a forward (noising) process that gradually adds Gaussian noise to the data, and a reverse (denoising) process learned via a neural network that removes the noise step by step.

Formally, the forward process defines a Markov chain:

$$q(x_t \mid x_{t-1}) = \mathcal{N}(x_t; \sqrt{1-\beta_t}\, x_{t-1}, \beta_t I) \tag{4}$$

$$q(x_t \mid x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}\, x_0, (1-\bar{\alpha}_t) I) \tag{5}$$

$$\bar{\alpha}_t = \prod_{s=1}^{t} (1-\beta_s) \tag{6}$$

Here, $\beta_t \in (0, 1)$ is a predefined variance schedule, and $\bar{\alpha}_t$ represents the cumulative product of noise terms. At inference time, the model learns the reverse process by estimating the noise added to the data at each time step:

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon, \quad \epsilon \sim \mathcal{N}(0, I) \tag{7}$$

$$p_\theta(x_{t-1} \mid x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta) \tag{8}$$

The neural network is trained to predict the added noise ϵ using the following simplified objective:

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{x_0, \epsilon, t}\left[\|\epsilon - \epsilon_\theta(x_t, t)\|^2\right] \tag{9}$$

This training objective enables the model to iteratively denoise inputs and generate high-quality synthetic samples by reversing the diffusion trajectory while avoiding issues like mode collapse common in GANs. Furthermore, while TimeGAN explicitly models the temporal structure, diffusion models implicitly preserve temporal coherence, offering a compelling alternative for time series generation. For a more in-depth discussion, we refer the reader to Ho et al. (2020b).

2.3. Time series image representations

Identifying an efficient representation is essential to leverage the effects of image-based GANs for time series synthesis. Transferring time series to images to infer statistical properties is a well-established and active research subject regarding generation and forecasting. All image representations for univariate time series construct a 2D encoding by performing a function f on the pairwise combination of time series x at two steps in time i and j. A time series of length S is transformed into a 2D matrix by

$$R_{i,j} = f(x_i, x_j) \quad \forall i, j \in \{1, 2, \ldots, S\} \tag{10}$$

while f can perform any operations on the two inputs. However, not all operations performed on the two inputs are suitable. The choice of f significantly impacts the representation's applicability and must be chosen carefully. Two criteria classify the functions applied to the time series return representations. The first criterion is invertibility. Invertible representations allow for encoding a time series in an image and retrieving the time series without any additional information. Non-invertible image representations often rely on a binary format, but even encodings in a continuous format may be non-invertible. Non-invertibility results from the function applied to the input pair x_i and x_j, or from a missing scale required for inversion, analogous to retransforming differenced values or a series of returns to their initial values. The diagonal of an image representation determines the invertibility. For the exemplary case of summation $f(x_i, x_j) = x_i + x_j$, the diagonal is defined by $R_{i,j} = x_i + x_j = 2x_i = 2x_j$. With this information, we can invert the representation, as opposed to the case of a differencing approach $R_{i,j} = x_i - x_j = 0$.

The second criterion is scale-invariance, which means that an image representation captures the characteristics of a time series regardless of its scale. A time series on different scales but with the same properties results in the same image. Note that the two criteria are not mutually exclusive and can be partially fulfilled. For the context of generating time series, a representation that encodes data in continuous values and allows for inversion while capturing properties in a scale-invariant way is a good candidate. Combining both criteria, they learn patterns across scales, allowing for easy inversion back to time series.

3. Related work

Three streams of literature are related to our work: research on image representations for time series, generative models, and methods for data augmentation. Below, we describe how each of the streams is affected by our contribution and discuss related approaches.

3.1. Time series image representation

We contribute to the current research on time series image representations in three ways.

First, we propose a continuous, deterministic image representation. Previous representations, such as the recurrence plot developed in the 1980s by Eckmann et al.

(1987), yield a binary encoding. They have found frequent applications, especially in chaos theory (Marwan et al., 2007) and system engineering (Shahverdy et al., 2020) to identify the state of an underlying system. Binary encodings do not support regression but can provide additional features in forecasting tasks (Li et al., 2020). However, reliance on ϵ, which defines a threshold value to indicate whether a point in time is a recurrence, is a drawback of binary encodings. The presence of non-stationarity makes choosing a fixed ϵ almost impossible. Our proposed method refrains from making an assumption on ϵ and instead relies on the logarithmic ratio of two points in time.

Another class of representations is Markov transition fields (Wang & Oates, 2015). They bin a time series into a distinct number of classes, calculate transition probabilities, and encode them in a two-dimensional space. The number of bins defines the dimension of the encoded image. Spectrograms also rely on probabilistic encoding. These are used similarly to a waveform, which processes audio signals and visualizes the spectrum of frequencies of a signal as it varies with time (Wyse, 2017). Each plot is then distinct for a certain sound or spoken word and can be used to classify audio signals.

Second, we devise an image representation that treats the image's diagonal differently from the off-diagonal to ensure invertibility. The only comparable invertible representations are the Gramian angular summation field (GASF) (Wang & Oates, 2015) and Fourier transform methods. Even though Fourier transform methods such as the short-term Fourier transform used by Zhang et al. (2023) in an exemplary application have proven successful, they are highly dependent on the choice of hyperparameters, such as the window size used for the transformation. This hinders immediate benchmarking, due to the large parameter space. Similar to the proposed XIRP, the GASF is a data-transformation method to convert one-dimensional time series data into a sequential set of images that retains the temporal features of the original data while adding temporal correlations. By representing each point in time by its polar coordinates and computing a cumulative sum, it treats off-diagonal values, which capture the intertemporal dynamics, the same as diagonal values. In the empirical part of the paper, we examine whether the separate treatment of diagonal and off-diagonal elements in the proposed XIRP offers an advantage over the GASF approach.

Third, the XIRP is more versatile than competing techniques. Binary recurrence plots, Markov transition fields, and spectrograms are non-invertible. Therefore, they support classification problems but are inadequate for point forecasts with continuous values. Unthresholded recurrence plots and GASFs support both tasks and have been applied to regression and classification (Barra et al., 2020; Zhang & Guo, 2020; Zhu et al., 2022). However, disregarding scale, they can only generate differenced time series and require a start value, which restricts their application in practice. Due to the scale-invariant temporal dynamics and scale-dependent diagonal, our representation is generally applicable for regression and classification. As part of the motivation for developing the XIRP, some representations are discussed more extensively in the upcoming section. In addition, Table 4 includes a summary of the prevalent methods and their characteristics.

3.2. Generative models

In the context of generative models, the paper offers two contributions.

First, embedding time series as images facilitates the use of state-of-the-art generative image models like GANs for time series synthesis, which simplifies the sampling process, reduces complexity, and raises sampling quality, as we show in the empirical part of the paper. In the past, generative models mainly relied on RNNs. Recently, transformer models have been introduced that act adversarially as generator and discriminator models (Li et al., 2022). With the advanced attention mechanism and design towards sequential data, the transformer architecture has already been successful in biological sequence analysis (Nambiar et al., 2020) and natural language processing (Gillioz et al., 2020). The results of Li et al. (2022) indicate competitiveness to a TimeGAN but solely rely on biometric motion data and two real-world datasets. Thus, TimeGAN remains a valid benchmark, and more thorough empirical validation is required for a conclusive substitution as a benchmark. The superior performance of TimeGAN compared to its competitors WaveNet (van den Oord et al., 2016) and RC-GAN (Esteban et al., 2017) results from additional networks that perform auto-encoding steps on a latent space as well as forecasting the latent space over time. The benefit of superior simulation results comes at the high cost of computing time, additional complexity, and hyperparameters. Generative models for time series relying on image representations have been proposed by Donahue et al. (2019), who transferred raw audio data to spectrograms and then used a generative model based on convolutional neural networks. For the application of time series synthesis, however, existing CNN-based generative models are outperformed by RNN-based alternatives and still suffer from the approximate invertibility of their representation. With the approach proposed here, there is no need for approximation, additional auto-encoding, or latent space supervision steps.

Second, we enable a large class of other generative models to benefit from the advantages of the XIRP. For example, apart from GANs, diffusion models from thermodynamics have been successfully applied to synthesize images (Durall et al., 2023). Instead of relying on an adversarial loss, their concept sequentially adds and removes noise in an input image. The sampling process is then analogous to the GAN framework, moving from a noisy image toward a synthetic instance. Since diffusion models are a relatively new concept, especially for time series settings, the research is still in an early stage. Another methodologically related stream of research examines the imputation of missing data in a probabilistic fashion (Alcaraz & Strodthoff, 2022; Park et al., 2022; Tashiro et al., 2021). These studies emphasize local coherence conditioned on an input series, which differs from our focus on unconditionally generating new time series.

A further merit of our approach may be seen in the fact that it could pave the way for generative models for multivariate time series synthesis, a field that is still in its infancy. In theory, our image-based approach generalizes

to the multivariate case. For example, WGAN-GP accepts stacked representations as joint input for the discriminator. We suspect, however, that in high-dimensional settings, the computational performance of the WGAN-GP is affected, since every time series would require a new channel. Even though we do not consider the multivariate setting in this paper, our approach might provide good results for lower dimensions. However, this still has to be proven empirically.

3.3. Data augmentation

The last stream of literature our paper contributes to is data augmentation. Our model complements existing augmentation methods with an adversarial learning method. Since the GAN samples from a noisy input, it opposes many techniques that transform the time domain by directly modifying the input time series, adding Gaussian noise or more complicated noise patterns such as spikes, step-like trends, and slope-like trends (Wen et al., 2021). Since it is non-parametric, it differs from the more advanced augmentation approaches that rely on decomposition and statistical generative model learning methods. Even though we also model a conditional distribution, our approach stands out from other conditional methods that typically involve modeling the dynamics of the time series with statistical models, e.g. Gaussian trees (Cao et al., 2014), by conditioning on noise instead of the previous point in time. Overall, the GAN-based approach complements the augmentation methods by a new category instead of being assignable to an existing one.

At this point, image-based generative processes often require an additional network to approximate the inversion of the representation. An invertible representation can simplify and improve the generative process via an adversarial approach with only two networks. An off-the-shelf WGAN-GP allows for quick training and may scale to higher dimensions. In sum, the WGAN-GP approach extends the set of data augmentation methods with the chance of enriching the training data, and supports the model's generalization capabilities.

The effects of data augmentation and its effects on enhancing regression and classification tasks have been investigated thoroughly (Iwana & Uchida, 2021; Liu et al., 2020; Semenoglou et al., 2023). However, image-based data augmentation techniques for time series are uncovered by existing literature. We aim to close this gap in the literature and examine whether the observed behavior in previous studies persists within the context of our research.

4. Methodology

This section briefly reviews existing image representations for time series before developing the ideas of the intertemporal return plot (IRP) and extended intertemporal return plot (XIRP).

Eckmann et al. (1987) introduced the first time series image representation. Their recurrence plot depicts points where the time series has approximately the same value. The representation is widely used and obtained as follows:

$$R^B_{i,j} = \begin{cases} 1 & \text{if} \quad \|x_i - x_j\| \leq \varepsilon \\ 0 & \text{otherwise,} \end{cases} \qquad \forall i, j \in \{1, 2, \ldots, S\} \quad (11)$$

with $R^B_{i,j}$ being an $S \times S$ matrix with indexes i and j. The representation is similar to a (binary) correlation matrix where each step in time corresponds to a feature, and S corresponds to the sequence length. A recurrence occurs if the absolute difference between the values at time i and j is smaller than a threshold ϵ. By definition, the diagonal d of $R^B_{i,j}$ is filled with recurrences $\forall i, j \in \{1, 2, \ldots, S\}, i = j \implies d_{i,j} = 1$. Consideration of the absolute difference also implies symmetry, $R^B_{i,j} = R^B_{j,i}$.

Recurrence plots incorporate significant statistical properties, including trends, jumps, and cyclic behavior. To extract such characteristics, recurrence plot quantification analysis (RQA) offers several proxies for the predictability of the underlying process or how long it remains in a particular state (Marwan et al., 2007).

Recurrence plots are powerful for time series analysis but require adjustments to serve as image representations for sampling synthetic time series. Encoding whether an absolute difference of time series values at indexes i and j exceeds ϵ, 'non-recurrences' can hypothetically take any value. This implies that binary recurrence plots are not invertible and cannot serve as input to a time series generation process.

We compute the difference between values at different time points or their ratio to ensure that a time series image representation is reversible. Differencing a time series stabilizes the mean and mitigates the impact of trend and seasonality (Sutcliffe, 1994). Generally speaking, differencing transforms a time series towards stationarity. This concept is popular in economics and quantitative finance and works well in many applications (Taylor, 2019; Tran & Kukal, 2022). To allow for a more granular approach than binary encoding, a naive recurrence plot (URP), sometimes termed a distance plot, can be obtained as follows (Iwanski & Bradley, 1998):

$$R^U_{i,j} = \|x_i - x_j\| \quad \forall i, j \in \{1, 2, \ldots, S\} \quad (12)$$

The URP conveys an idea of the magnitude by which x_i and x_j differ and thus adds additional information to the representation. Even though it improves upon R^B, the approach has three drawbacks. First, the differencing approach is acceptable if the feature scales do not change drastically. In cases where the values are naturally in persistent bounds, differencing is legitimate and likely to return good results if the time series evolves. However, if the time series evolves and the mean of the feature jumps to a higher level or exceeds its previous bounds, the relations discovered for past differences on an absolute level will no longer hold. Therefore, the representation $R^U_{i,j}$ is not scale-invariant, which decreases the chance of obtaining a robust representation across scales. Second, the difference plot does not contain the scale of the feature, which vanishes by subtracting. A missing scale makes it impossible to put the changes into perspective, leading to the third drawback. The absolute values make

it impossible to use the representation in a generative setting, since the initial sign of the change is unknown. In case one would calculate an alternative representation of $\hat{R^U}_{i,j} = \|x_i - x_j\|$, the diagonal contains $\hat{R^U}_{i,i} = 2x_i$ and would be invertible but still not scale-invariant.

The alternative approach of using (logarithmic) ratios is frequently used in financial applications. The log transformation has a similar effect on the time series as differencing. However, calculating returns has the advantage of being invariant to scale and is especially useful when the time series evolves. Further, log returns have the desirable statistical characteristics of symmetry and additivity (Wilmott, 2006, p. 295). Motivated by the benefits of a scale-invariant representation, we propose the intertemporal return plot $R^I_{i,j}$, defined by

$$R^I_{i,j} = log\left(\frac{x_i}{x_j}\right) \quad \forall i,j \in \{1,2,\ldots,S\} \tag{13}$$

An advantage of R^I over R^B is the absence of ϵ, the correct specification of which is complicated and may change over time. Further, R^I indicates the direction compensating the limitations of R^U while remaining scale-invariant. However, the R^I also allows inverting the time series without a specific start value. A careful look at R^I reveals that in case $i = j \implies R^I_{i,j} = 0$, which leads to a diagonal filled with zeros. This inefficiency motivates the formulation of the XIRP, which complements the intertemporal return structure by adding extra information to the encoding. Thus, Eq. (3) evolves to

$$R^X_{i,j} = \begin{cases} log\left(\frac{x_i}{x_j}\right) & \forall i,j \in \{1,2,\ldots,S\} \wedge i \neq j \\ x_i & \forall i,j \in \{1,2,\ldots,S\} \wedge i = j \end{cases} \tag{14}$$

The previously unused diagonal of the IRP is filled with additional information. R^X now has all relevant characteristics regarding scale-invariance and invertibility required for a generative model.

Another approach that provides both scale-invariance and invertibility is the Gramian angular summation field (GASF) concept. It combines the previously mentioned benefit of summation proposed for R^U with bounded feature scales. The GASF representation has already been tested extensively in various applications, from forecasting sun flares (Chen et al., 2020; Nagem et al., 2018) to stock prices (Liang et al., 2022). It transfers a [0, 1] scaled time series x for a point t in time to

$$\phi_t = arccos(x_t) \tag{15}$$

which allows for calculating the GASF for two points in time via

$$R^{GASF}_{i,j} = cos(\phi_i + \phi_j) \quad \forall i,j \in \{1,2,\ldots,S\} \tag{16}$$

According to Wang and Oates (2015), it provides a way to preserve the temporal dependency, since time increases as the position moves from top-left to bottom-right in $R^{GASF}_{i,j}$, and the main diagonal allows us to reconstruct the time series. Even though it scales the input beforehand, it is not scale-invariant, since multiplying the scaled input by a constant returns a different GASF in the off-diagonals. This is considered a drawback, since the dynamics remain unchanged but are encoded differently in the GASF. In practice, the GASF, IRP, and XIRP require a scaled time series before being applied. Afterwards, descaling allows us to obtain the values at the initial scale.

A last representation introduced mainly for benchmarking purposes is the naive representation defined by

$$R^N_{i,j} = x_i \quad \forall i \in \{1,2,\ldots,S\} \tag{17}$$

Technically, the representation concatenates the time series columns S times, returning an image where each row has the same value. The naive representation removes all additional information that could originate from an advanced representation. At this point, there are two reasons why we need a naive baseline representation to compare GASF and XIRP to the results of TimeGAN. First, we would like to identify the benefit of each representation for measures of similarity and predictive ability compared to the naive representation. Second, we are comparing two inherently different models in the benchmark section. TimeGAN has five loss functions compared to a WGAN-GP with one loss function. We use the naive representation to reduce the possible benefit of a superior representation, so neither TimeGAN nor WGAN-GP has proprietary information.

Fig. 1(a) displays a random time series of $S = 20$ steps with a positive trend. For the recurrence plot in Fig. 1(b), the dots mark the recurrence and cluster around the diagonal, which is zero by definition. Clusters around the diagonal indicate a drift in the time series, since recurrences only appear in short periods as the time series evolves. The IRP and XIRP look similar with the difference of the diagonal, which is 0 in the IRP and contains the initial time series values from Fig. 1(a) in the diagonal. Both plots show a significant return at the beginning of the time series and smaller ones in the later course. The naive representation in Fig. 1(e) shows stripes resulting from the time series' column-wise concatenation. The GASF displayed in Fig. 1(f) shows a contrasting pattern assigning smaller values to the lower-left corner and higher values to the upper-right, since it refers to the polar encodings and not its returns. The naive representation completes the set of candidates for the subsequent benchmark. Linking the concepts of GAN and XIRP/GASF, we can sample and extract the time series from the diagonal and the temporal dependencies in the off-diagonal.

There are multiple ways to invert the underlying time series from both representations. Since the diagonal contains the initial values, they are extractable without considering the off-diagonal. Although this inversion is most accurate numerically, it would neglect parts of the encoded information and lead to inferior simulation results. Thus, we propose extracting the diagonal in the first step and then using the diagonal as a baseline time series to calculate variants via the off-diagonal returns. Once all n series are inverted, we can either average all series, which we refer to as inversion by mean (IM), or we can randomly select an instance, which we label inversion by random column (IRC). Note that both IM and IRC are approximate inversions.



(a) AR(1) TS (b) Recurrence plot with $\epsilon = .2$ (c) Intertemporal Return Plot (IRP)

(d) Extended IRP (XIRP) (e) Naive Representation Plot (f) Gramian Angular Summation Field (GASF)

Fig. 1. Visual representations of an AR(1) together with additional time series encodings.

A simple example can illustrate the approach with the help of Fig. 1. In the case of the naive representation, the IM method averages over each row of the plot, obtaining one value for each line. In the case of IRC, a column is selected randomly.

5. Data

In this section, we provide an overview of the synthetic and real datasets utilized in our experiments, detailing their properties and the rationale behind their selection. For our experiments, we use publicly available real-world and simulated time series data. In total, 100 time series from nine datasets test the generative capabilities of each model and representation. A limit of 1000 observations is set for each time series to speed up training and facilitate comparability between the scores among datasets.

The first simulated datasets are the Sine and Noisy Sine datasets. They consist of nine time series of sine curves with varying shifts and levels of noise. They are incorporated to check how well the models and representations handle (noisy) cyclic behavior. Another critical aspect of a generative model is its handling of randomness and Markov properties, both incorporated in Brownian motion. The trajectory undergoing Brownian motion is unpredictable and exhibits erratic movements. It is a stochastic process where future behavior depends only on the current state and is independent of its past trajectory. Each change in direction is independent of previous changes. To test if a model successfully captures this behavior, we construct a Brownian Motion dataset with time series of varying coefficients for σ ranging from $\sigma = 0.9$ to $\sigma = 0.99$ and $\mu = 0$.

Evaluating the performance of generative models under disruptive circumstances is the next point of interest. In reality, we often observe time series with jumps in the context of news releases, economic shocks, or other sudden changes in market conditions, which a Merton jump process can model. It includes a jump intensity parameter λ that governs the frequency and magnitude of the jumps and the volatility of the diffusion process σ. To test the capabilities of models and representations concerning jump processes, the Merton Process dataset contains 10 time series with varying values for λ and component σ. Another characteristic commonly observed

in real-world data concerns heavy-tailed distributions, implying a slower decay than distributions with finite moments. The heavy tail suggests that rare events with large magnitudes are more likely to occur under a normal distribution than expected. We test a process where the change is governed by a power law function in which a slight increase is typical while more significant increases are rare. In the corresponding experiments, α is in the range of [0.1, 0.9].

Next, we consider the real-world datasets, starting with the UCI Appliances energy prediction (Dua & Graff, 2017) dataset, characterized by noisy periodicity. The data consists of multivariate, continuous-valued measurements, including numerous temporal features measured at close intervals. By contrast, sequences of the stock price dataset are continuous-valued but aperiodic. We use the first 1000 trading days of historical Google stock data from 2004 to 2008, including volume and high, low, opening, closing, and adjusted closing prices. The Air Quality dataset contains the responses of gas sensors near an Italian city containing hourly response averages recorded along with gas concentrations (Dua & Graff, 2017). The dataset combines characteristics of the synthetic datasets and includes a total of 13 time series with different properties in terms of periodicity and jumps. The Bike Share dataset combines the seasonal patterns with irregular jumps for time series like weather conditions, precipitation, day of the week, season, hour of the day, and rentals by hour. The exact number of time series n for each dataset can be obtained from Table 1.

6. Experiments

In the following, we first revisit our research questions and describe how each experiment contributes answers. Next, we outline how qualitative and quantitative experiments cover different aspects of time series synthesis quality and introduce an evaluation scheme for the obtained results.

The key questions to answer by empirical experimentation are as follows. Which image representation is better suited to capture temporal dynamics? How does the proposed pipeline of time series image representation, image generation, and image inversion compare to state-of-the-art generative time series models? And are image-based time series representations with their respective generative models effective in enhancing generalization?

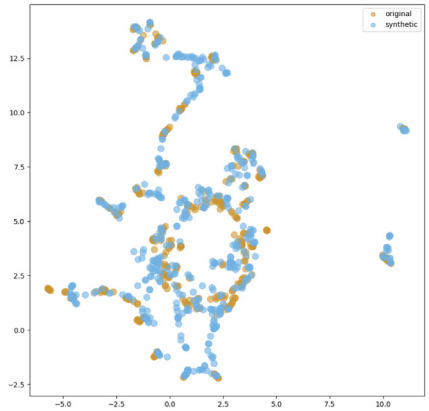
To answer these questions, we conduct a series of experiments. First, a visual inspection compares synthetic and original time series using clustering techniques to address the challenges of direct high-dimensional comparison. Next, we assess predictive ability by testing how accurately models trained on synthetic data can predict real data, offering insights into the quality of the synthetic data. Additionally, a test for forecast enhancement evaluates whether synthetic data can improve model performance on future, unseen data, thereby demonstrating its practical utility in training. We also conduct a similarity of moments test to determine how well the generative models capture the statistical properties of the original data, which is crucial for ensuring the realism of synthetic data. Finally, a discriminative test measures the ability of a classification model to distinguish between synthetic and original data; higher classification errors indicate a closer resemblance between the two, reflecting higher-quality synthesis. Collectively, these tests provide a comprehensive evaluation of the model's capabilities and the utility of synthetic data.

In the following, we introduce a series of evaluation tests used to assess the quality of synthetic time series. For each test, we briefly summarize the most important findings to combine high-level insights with a structured line of argument. A detailed analysis of all results is provided later in the empirical results section.

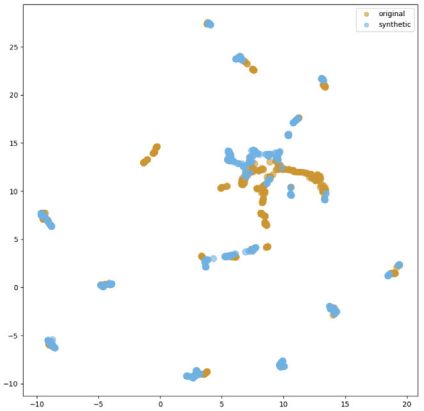
6.1. Visual inspection

For the visual inspection, we first calculate image representations for a set of real-time series. Next, we train WGAN-GP to simulate synthetic XIRPs, GASFs, and naive representations with a sequence length of $S = 28$. Since the simulation is unconditional, directly comparing the synthetic and original time series is not feasible. To enable visualization nevertheless, we apply a clustering algorithm to facilitate comparisons of synthetic and original time series in a two-dimensional space. A single dot represents an original or synthetic time series, depending on its color. Elements located closely in the two-dimensional space will likely have similar characteristics in the higher-dimensional space. The method used for clustering is the uniform manifold approximation and projection (UMAP) algorithm (McInnes et al., 2020), which interprets the d time steps as an individual feature. Ali et al. (2019), Yoon et al. (2019) use a similar approach for clustering time series. In case original and synthetic elements differ, the UMAP algorithm forms separate, homogeneously colored clusters with little distance between the elements inside each cluster and large distances between the individual clusters. Coloring the elements in the plot visualizes their class membership of either original (brown) or synthetic (blue). The algorithm returns heterogeneous clusters of mixed red and green dots if the elements are similar. In the first part of the empirical results, we discuss three arbitrarily chosen synthesis results from WGAN-GP and TimeGAN. This is for illustration. Recall that the total number of combinations of XIRPs, GASFs, and naive representations with recovery methods IRC and IM accumulates to $n_{comb} = 600$. Hence, it is impossible to analyze all samples visually. For every UMAP plot, we randomly chose 500 original and synthetic elements.

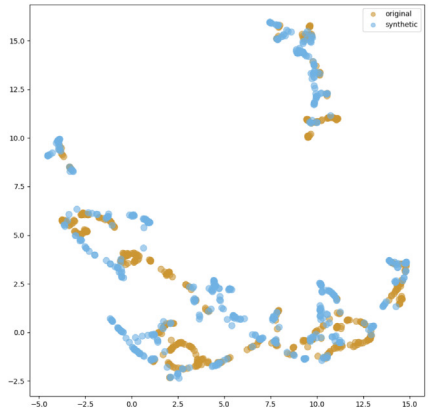
The first example displayed in Fig. 2(a) shows original and synthetic elements generated by WGAN-GP on the



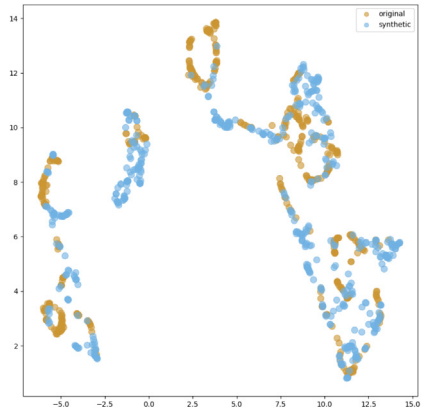
(a) WGAN-GP on bike share *cnt* (XIRP, IRC)



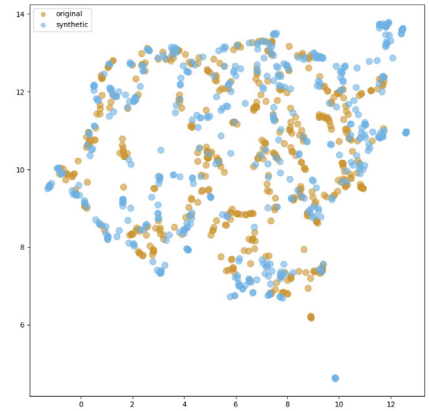
(b) TimeGAN on bike share *cnt*



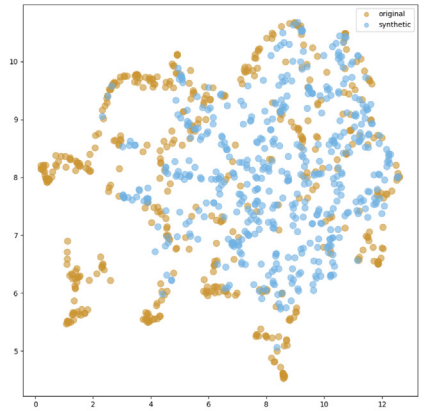
(c) WGAN-GP on stock data *Adjusted Close* (XIRP, IRC)



(d) WGAN-GP on stock data *Adjusted Close* (GASF, IRC)



(e) WGAN-GP on *hum* of bike share (XIRP, IRC)



(f) Diffusion on *hum* of bike share

Fig. 2. UMAP projections for real and synthetic time series (cross-model, cross-representation comparison); closer dots indicate higher similarity between real and synthetic data.

count of total rental bikes (*cnt*) of the Bike Share dataset. The original elements marked in brown predominantly cluster in the left half, consisting of a circular shape in the center and two curved shapes to the left and top. Further, a small dense cluster forms to the upper-right of the circular shape. A few isolated original synthetic points are spread on the right. The synthetic elements distribute well between the original elements and come close to their shape. Surprisingly, the synthesis covers even the remote real elements on the right. Synthetic elements below the circle of real instances bridge the gap between real and synthetic instances, proposing new plausible shapes. From a visual inspection, synthetic elements augment the data well.

Fig. 2(b) displays the result for *cnt* for the TimeGAN model. Original instances mainly populate the center of the UMAP. We also observe more separate clusters mixed with synthetic and original instances. The visual inspection suggests that the TimeGAN model performs worse for this dataset than the WGAN-GP. Compared to 2(a), the shapes differ significantly, due to the random choice of the instances and the difference in the synthesis result. Since the UMAP uses randomness for fast approximation and solving optimization problems, different runs of UMAP can produce different results.

Also, the synthetic elements are spread relatively evenly among the original instances in the new plots, reflecting strong augmentation quality. In Fig. 2(c), the WGAN-GP model applied to stock data (*Adjusted Close*, XIRP, IRC) demonstrates good alignment, with synthetic points (blue) distributed across most regions occupied by original data (orange). However, a slight concentration of synthetic elements appears in the top-right cluster, while the lower-left remains more sparsely populated.

In Fig. 2(d), WGAN-GP with the GASF representation shows a distinct 'beads on a string' structure, where synthetic and original samples form heterogeneous clusters. Although the alignment appears strong overall, some regions, particularly in the center, are more heavily populated by original data, suggesting minor underrepresentation in the synthetic data. Fig. 2(e) presents the WGAN-GP results on the Bike Share humidity data (*hum*) using the XIRP and IRC encoding. The synthetic and original samples mix well, forming a densely populated central region. Small synthetic-only clusters appear in isolated corners, such as the upper right, but these may result from sample selection and do not necessarily indicate poor generation quality.

By contrast, Fig. 2(f) shows the results from the diffusion model on the same *hum* series. While there is substantial overlap between the original and synthetic points, several larger regions remain uncovered by synthetic data, particularly around the edges. This may indicate a systematic issue in capturing the full distribution, as opposed to random gaps from sampling. Still, most original instances have nearby synthetic counterparts, suggesting partial but uneven alignment. For completeness, additional side-by-side plots of real and synthetic time series were included to complement the UMAP visualizations. While more challenging to interpret and limited to a subset of 100 series, they offer further insights into the temporal structure and variability alignment between real and synthetic data. See Fig. 7 in the Appendix for a collection of these time series plots.

Regarding the first question—which image representation is better suited to capture temporal dynamics—we cannot make a definitive statement at this stage. For the XIRP and GASF, we observe successful (and imperfect) synthesis results. Both representations catch the dynamics well and deliver good results according to the UMAP algorithm. To answer the second research question—whether an image-based model returns competitive results when benchmarked against state-of-the-art time series generative models—a first impression suggests that the characteristics of the time series generated by WGAN-GP are comparable to those obtained by TimeGAN, and slightly favorable compared to a diffusion model.

6.2. Predictive ability

For the experiment, we require quantifiable and comparable measures of model performance. The goal is to measure the predictive ability of each model, representation, and inversion method. The first backtest trains a downstream model solely on synthetic data to predict the original data. The intuition is that if the synthetic data are close to the original data, after being trained on synthetic data, the trained model should be able to predict the original data. We train a gated recurrent unit (GRU) on synthetic data sequences from $t = 0$ up to $t = d - 1$ and predict each sequence's dth element. We repeat the process $k = 10$ times to obtain reliable results and apply early stopping as a regularization technique before calculating the MAE on a test set of original data. We then average the MAE over k folds. The lower the MAE, the better the model synthesizes the data.

Table 1 summarizes the scores for each model, metric, dataset, and representation, with average scores on the left side and the best result highlighted in boldface. On the right side of the table, we report the number of best scores. Note that we use IRC to convert the results from image-based models.

At a high level, results for predictive ability show that GASF and XIRP consistently outperform the Naive representations. Among models, WGAN-GP demonstrates a clear advantage over TimeGAN and Diffusion, particularly when paired with GASF or XIRP. These combinations yield the lowest S_P scores across a wide range of datasets, suggesting that they capture the underlying structure of the data more effectively than other approaches. Multiple datasets highlight the strength of GASF, which achieves the lowest predictive scores and demonstrates a strong capacity to model periodic and noisy signals. As the data become more complex—such as in the Brownian Motion and Merton Process datasets—GASF and XIRP remain strong performers, indicating their ability to encode

stochastic dynamics. On the Energy dataset, XIRP slightly outperforms GASF, suggesting its advantage at handling intricate real-world data. For the Stock Data and Air Quality datasets, performance is more evenly distributed, although XIRP and TimeGAN often appear among the top representations.

Overall, the results provide a clear performance hierarchy. Image-based representations, particularly GASF and XIRP, offer robust and consistent predictive performance across synthetic-to-real generalization tasks. WGAN-GP, when combined with these representations, emerges as the most effective generative model in terms of predictive accuracy.

6.3. Forecast enhancement

The second forecast enhancement test quantifies how synthetic data reduce forecast errors on unseen data. While the first test allows a relative comparison of the models and representations, this test provides sufficient evidence of the extent to which synthetic data help the model better generalize. To test for forecast enhancement, we train a GRU on $n_{ori} = 1000$ original observations and compare its performance to a model trained on $n = 2000$ observations, $n_{ori} = 1000$ original and $n_{syn} = 1000$ synthetic ones. The first n_{ori} observations constitute the training and validation set, and the last n_{ori} form the test set. The gaps between validation and testing are large enough so that neighboring periods are embargoed to prevent data leakage and ensure the evaluation's integrity where future information is not available at the time of prediction (De Prado, 2018, p. 107). Analogously to the first test, synthetic data sequences range from $t = 0$ up to $t = d - 1$ and predict each sequence's dth element. We construct an augmentation score S_A which captures the improvement with respect to the root mean squared error (RMSE). The S_A divides RMSE on original and synthetic data by the result on original data. Thus, the metric is defined as $S_A^{RMSE} = \frac{RMSE_{ori+syn}}{RMSE_{ori}}$. S_A values below (above) 1 thus indicate an improvement (deterioration) in the metric. The forecast enhancement test only shows whether a prediction improves. A lower score does not mean that the forecast is better in absolute terms. Thus, cases that achieve poor results on real data but good quality on synthetic and real data achieve a better score than those that consistently achieve good results and only deliver a small percentage improvement. This motivates the introduction of a third test, a distributional test, which evaluates the effectiveness of different representations and models at generating samples that accurately reflect the statistical properties.

In terms of results, the Naive representation consistently outperforms others in terms of reducing forecast errors, with the lowest S_A values across a wide range of datasets. For example, it achieves exceptionally low scores on synthetic datasets like Sine (0.22) and Noisy Sine (0.125), suggesting strong generalization in simple periodic and noisy environments. Surprisingly, this advantage extends to complex datasets as well—Energy (0.275), Stock Data (0.313), Air Quality, and Bike Share—demonstrating the robustness of simple representations when combined with effective model training. Across all evaluations, WGAN-GP proves to be the most effective generative model for reducing forecast errors, achieving the best S_A scores in 73/100 cases and vastly outperforming TimeGAN (11/100) and Diffusion (16/100). In summary, simple representations and robust generative models offer substantial forecast enhancement, with WGAN-GP (Naive) emerging as a particularly effective combination.

6.4. Similarity of moments

Since the generated sequences comprise multiple time steps, the intertemporal metric space is vast, and an exhaustive analysis is impractical. Therefore, we focus on analyzing the first four moments of the mean and standard deviation across sequences for each time series. A small difference in the moments suggests that the generative model effectively captures the underlying data-generating process. Conversely, discrepancies in moments may indicate areas where the generative model fails to replicate the original data accurately.

Table 3 provides a comparative ranking of the models and representations. The WGAN-GP with XIRP representation emerges as the most effective overall, achieving the lowest average rank.

6.5. Discriminative test

In a fifth discriminative test, we use each generative model to synthesize data and let a GRU discriminate between original and synthetic data. In this test, lower (higher) classification accuracy indicates high (low) similarity between the synthetic and original data. Thus, the test aims at judging the similarity between synthetic and original elements. A higher binary classification error signals a higher quality of generated data. Again, we use early stopping and a test set for all $k = 10$ iterations before averaging the binary cross-entropy error.

Exploring the S_D metric, which is understood to measure the mean discriminative score (where higher scores indicate better performance), reveals the proficiency of different models at distinguishing patterns or features specific to each dataset. This metric is pivotal in evaluating how well models can differentiate between various data characteristics and patterns.

In our tests, the performance on the S_D score is more balanced than on S_A and S_P. The Naive representation often achieves top S_D scores, even outperforming more complex encodings on both synthetic and real-world datasets. Its strong performance on datasets such as Energy Data secures it the top rank (25/100), closely followed by TimeGAN (22/100). Meanwhile, GASF and XIRP show competitive or superior performance. This diversity of strengths among representations demonstrates that no single approach dominates across all contexts. The next section presents a detailed breakdown of S_D performance across datasets and models, revealing how these trends play out at a finer level of granularity.

6.6. Time series characteristics and representation success

In our study, we investigate the influence of time series characteristics on the performance of different representations in synthesis. This analysis is crucial for selecting an appropriate representation based on the characteristics of the target time series. By calculating correlation coefficients between various time series characteristics—including standard deviation (σ), the augmented Dickey–Fuller (ADF) test statistic for stationarity, the trend coefficient (x_0) from linear regression, the autocorrelation function (ACF) at different lags, and the jump intensity (JI)—we uncover how these statistical properties of time series correlate with performance metrics S_P and S_D across various representations.

This analysis reveals that the GASF representation tends to perform well at predictive tasks (S_P) when applied to moderately variable and slightly non-stationary time series, while the XIRP shows similar but slightly weaker patterns. In contrast, the Naive representation shows a unique strength: although it generally correlates negatively with S_P, it excels when abrupt changes are present, as reflected in its strong positive correlation with jump intensity. For discriminative performance (S_D), both GASF and XIRP representations show decreasing effectiveness as variability and trend increase, indicated by their negative correlations with standard deviation and trend coefficients. The Naive representation, however, exhibits an interesting inverse pattern: it performs best when autocorrelation is strong, even though it struggles with other statistical properties. This suggests that Naive is particularly useful for identifying recurring patterns and detecting anomalies that hinge on the raw sequence values.

These findings guide a more informed selection of representations. Time series with persistent autocorrelations may benefit from the XIRP encoding, while those featuring sudden shifts or requiring anomaly detection may be better served by the Naive approach. Ultimately, the choice of representation should reflect both the statistical profile of the time series and the specific objectives of the synthesis task. Our analysis provides the basis for such decisions, aligning representation choice with data characteristics to improve model outcomes.

6.7. Inversion methods and computation times

The upper section introduces two inversion methods of which the IRC results from observing the harmful effects of IM on the training data. Visual inspections of synthetic time series obtained by IM and IRC have shown that IM returns smoother trajectories, which differ from the input time series and are thus easy to detect in the backtests. Fig. 4 compares the performance for each representation and inversion method for each S_D and S_P. A look at Fig. 4(a) shows that the IM inversion obtains a lower score on the discriminative metric. Since a high discriminative metric is best, we see that IRC improves the score for each representation.

A detailed analysis reveals that IM results in lower discriminative scores, which is expected given its smoother outputs. In contrast, IRC consistently enhances discriminative performance across representations, with especially notable gains in S_D, where improvements range from approximately 4.8% in the naive case to over 360% for GASF and XIRP. Predictive performance also improves substantially for GASF and XIRP—exceeding 60%—while the naive representation slightly deteriorates. These findings underline the benefits of IRC, particularly when combined with structured representations, in generating synthetic data that better mimic the original sequences. Computation time is another critical factor when selecting models. Using an Nvidia Tesla T4 GPU, WGAN-GP and Diffusion trains significantly faster than TimeGAN, completing training and backtesting in roughly 10 to 12 min per time series, compared to around 25 min for TimeGAN. While these results are promising, they come with the caveat that the TimeGAN implementation was not optimized for GPU usage. Moreover, hyperparameters such as the batch size, learning rate, and number of epochs were held constant at TimeGAN's defaults, ensuring a fair comparison but also potentially limiting the full capacity of WGAN-GP. Still, the simpler architecture of WGAN-GP and Diffusion with only one loss function—and in the case of WGAN-GP, two networks—demonstrates competitive performance. This supports the conclusion that improvements observed under IRC are not solely due to model architecture but are in part driven by the enhanced inversion process.

7. Empirical results

After the high-level overview, we now turn to a more detailed analysis, using exact comparisons and a ranking system to better assess the model and representation performance. A look at Table 1 reveals that all models obtain the best scores on some of the datasets. Across the nine datasets, TimeGAN achieves the best S_D score in 4/9 cases but never achieves the best S_P score. WGAN-GP obtains the best S_P score on 5/9 datasets and the best S_D score on 2/9 datasets using GASF. Using the XIRP representation, the WGAN-GP scores best on 5/9 S_P and 3/9 S_D scores. Even training on a naive representation achieves the best result on some occasions (e.g. lowest S_P score on 1/9 and highest S_D score on 2/9 datasets). For S_A, the naive representation scores best on 7/9, leaving 2/9 for the TimeGAN model.

Considering the individual time series level, the best-performing model regarding the average score usually has the highest number of wins for the dataset. A striking exception is S_D on Energy Data. Even though the average discriminative scores for TimeGAN ($S_D = 0.591$) and the WGAN-GP trained on the naive representation ($S_D = 0.590$) are close, the latter wins on 19/27 scores. This imbalance indicates that the naive model generally performs better, but in a few examples it failed to create sufficiently similar instances, which has a strong negative impact on the average. In total, the image-based methods won 21/27 S_D, S_P, and S_A scores. Regarding S_P, performance differences among the representations are substantial. The WGAN-GP trained on R^N and TimeGAN score similarly, winning only 6/100 and 8/100 time series, respectively,

Table 1
Benchmarking results with mean predictive score S_P (the lower the better), mean discriminative score S_D (the higher the better), and S_A (the lower the better).

Data	n	Metric	TimeGAN	WGAN			Diffusion	# Best Scores				
			TS	GASF	XIRP	Naive	TS	TimeGAN	GASF	XIRP	Naive	Diffusion
Air Quality	13	S_D	0.315	0.630	0.629	**0.631**	0.249	0	3	3	3	**4**
			(0.022)	(0.017)	(0.019)	(0.016)	(0.084)					
...	...	S_P	0.299	0.248	0.248	**0.247**	0.521	**4**	3	**4**	2	0
			(0.048)	(0.106)	(0.106)	(0.106)	(0.236)					
...	...	S_A	**0.901**	1.013	0.962	0.998	1.000	4	0	1	3	**5**
			(0.0833)	(0.1185)	(0.097)	(0.078)	(0.004)					
Bike Share	7	S_D	0.508	**0.667**	0.660	0.664	0.115	0	**3**	2	2	0
			(0.026)	(0.017)	(0.015)	(0.015)	(0.006)					
...	...	S_P	0.730	0.663	**0.656**	0.663	0.667	1	0	**3**	1	2
			(0.008)	(0.009)	(0.009)	(0.106)	(0.016)					
...	...	S_A	**0.924**	0.989	0.975	0.974	1.000	0	1	**4**	0	2
			(0.037)	(0.123)	(0.051)	(0.106)	(0.103)					
Brownian Motion	10	S_D	0.378	0.314	0.334	**0.556**	0.530	0	1	1	2	**6**
			(0.031)	(0.048)	(0.054)	(0.026)	(0.027)					
...	...	S_P	0.248	**0.095**	0.095	0.250	0.245	0	**5**	4	1	0
			(0.030)	(0.010)	(0.011)	(0.037)	(0.036)					
...	...	S_A	1.163	0.751	1.243	**0.175**	1.008	1	0	**5**	1	3
			(0.512)	(0.4021)	(0.323)	(0.299)	(0.035)					
Energy Data	27	S_D	**0.591**	0.450	0.445	0.590	0.461	7	2	3	**14**	1
			(0.026)	(0.053)	(0.056)	(0.026)	(0.021)					
...	...	S_P	0.352	0.092	**0.092**	0.347	0.441	0	**14**	12	1	0
			(0.015)	(0.015)	(0.019)	(0.074)	(0.056)					
...	...	S_A	0.641	0.946	0.884	**0.274**	1.005	2	4	1	**18**	2
			(0.109)	(0.198)	(0.221)	(0.281)	(0.016)					
Merton Process	10	S_D	**0.462**	0.456	**0.462**	0.416	0.290	**5**	1	3	1	0
			(0.019)	(0.047)	(0.052)	(0.029)	(0.018)					
...	...	S_P	0.270	**0.102**	**0.102**	0.278	0.281	1	4	**5**	0	0
			(0.006)	(0.006)	(0.006)	(0.017)	(0.017)					
...	...	S_A	0.675	1.061	1.070	**0.107**	1.010	1	2	2	**4**	1
			(0.006)	(0.006)	(0.006)	(0.017)	(0.018)					
Noisy Sine	10	S_D	0.412	0.402	**0.423**	0.532	0.361	2	2	**3**	1	2
			(0.026)	(0.047)	(0.055)	(0.028)	(0.031)					
...	...	S_P	0.309	**0.105**	**0.105**	0.329	0.323	1	**5**	4	0	0
			(0.027)	(0.015)	(0.015)	(0.040)	(0.052)					
...	...	S_A	0.7182	1.0437	1.202	**0.1250**	1.0120	1	1	0	**6**	2
			(0.0345)	(0.0585)	(0.045)	(0.367)	(0.274)					
Power Law	9	S_D	**0.670**	0.441	0.442	0.625	0.583	**7**	1	0	1	0
			(0.015)	(0.075)	(0.064)	(0.037)	(0.020)					
...	...	S_P	0.377	**0.125**	**0.125**	0.360	0.347	0	**5**	3	0	1
			(0.105)	(0.093)	(0.093)	(0.106)	(0.095)					
...	...	S_A	0.4990	0.8940	0.947	**0.137**	1.000	1	1	2	**5**	0
			(0.08)	(0.054)	(0.037)	(0.066)	(0.002)					
Sine	9	S_D	0.584	**0.670**	0.669	0.523	0.094	1	**5**	1	0	2
			(0.020)	(0.011)	(0.014)	(0.036)	(0.016)					
...	...	S_P	0.469	**0.079**	0.079	0.4298	0.444	0	**6**	2	1	0
			(0.140)	(0.005)	(0.005)	(0.121)	(0.124)					
...	...	S_A	0.702	1.0312	0.945	**0.220**	0.997	0	1	1	1	**2**
			(0.046)	(0.107)	(0.246)	(0.194)	(0.598)					
Stock Data	5	S_D	**0.593**	0.527	0.510	0.549	0.067	1	1	**2**	1	0
			(0.022)	(0.060)	(0.051)	(0.026)	(0.016)					
...	...	S_P	0.360	0.160	**0.100**	0.362	0.765	1	0	**4**	0	0
			(0.048)	(0.034)	(0.026)	(0.052)	(0.000)					
...	...	S_A	0.728	0.983	0.992	**0.313**	1.000	1	0	1	**3**	0
			(0.013)	(0.154)	(0.382)	(0.0156)	(0.000)					

while the Diffusion model is only best in 3/100 cases. The XIRP scores better with 41/100 only surpassed by GASF, winning 42/100 time series. On Energy Data, GASF wins 14/27 time series, with $S_P = 0.092$, slightly outperformed by XIRP ($S_P = 0.092$) despite winning fewer series (12/27). We attribute this to a few particularly low-performing samples. The WGAN-GP model regularly yields similar results for GASF and XIRP. The predictive score, e.g. for the Merton Process, Noisy Sine, and Power Law, match up to the fourth decimal place, sharing the small values.

The S_A score indicates that for every dataset we find a representation able to reduce the forecast error. With the majority of scores being smaller than 1, we can state that the synthetic data help the model to generalize better compared to only real data. Surprisingly, all models except the Diffusion model achieved average S_A scores below one. In contrast, the Diffusion model consistently exhibited average S_A scores above one. However, it performed well regarding the absolute number of best S_A scores across individual cases, suggesting that the Diffusion model excels in certain instances. Its overall performance, however, is dragged down by poor results on specific parts of the dataset, thereby inflating its average score. The exceptionally low values for the naive representation and TimeGAN originate from higher forecasting errors in the first instance, as represented in S_P, which lead to a large fractional reduction in the following.

It becomes evident that the representations and models perform differently regarding the scores. While the naive representation shows the best results for S_D, it performs second worst, closely followed by TimeGAN, on S_P. Approximately, GASF and XIRP win twice as many time series on S_D compared to S_P, opposing the performance of TimeGAN. While GASF and XIRP show good predictive scores, their forecast enhancement captured by S_A is relatively small in comparison.

Regarding the cross-representation comparison, the tests do not identify a superior representation across all scores. It rather depends on the application of the model. If the goal is to find the most similar instances, choosing a WGAN-GP using a naive representation is advisable. If, on the other side, the goal is to synthesize data to augment the existing data for training, then the choice falls on a WGAN-GP using GASF or XIRP as a representation. However, the simpler WGAN-GP model outperformed the TimeGAN model in all three, delivering significantly better results with the naive approach on S_D and S_A and GASF on S_P. Since our naive representation ensured the same level of information, we can state that the WGAN-GP is competitive with the TimeGAN model.

Now that the average metric score and the number of best scores are known, we evaluate the ranking of each model. We calculated a ranking for S_D and S_P on each time series and assigned ranks from 1 (best) to 5 (worst). Comparing the average rank puts the performance of each model and representation into perspective. Regarding S_D, we find that the WGAN-GP model has an average rank of 3.21 while the TimeGAN has an average rank of 2.96 and Diffusion scores best with an average rank of 2.47. For S_P, however, the TimeGAN and Diffusion models have average ranks across all time series of 3.49 and 3.7 while the WGAN-GP has an average rank of 2.6. For S_A, the WGAN-GP and Diffusion models have average ranks of 2.76 and 3.63 across representations, while TimeGAN scores an average rank of 2.56. The score of the WGAN-GP is drawn down by the inferior results of the GASF and XIRP, while the naive representation alone achieves an average rank of 2.37. While the Naive representation is strong in terms of average ranks of S_A (2.37), averaging the scores with GASF (2.99) and XIRP (2.92) leads to an average that is slightly higher than that of TimeGAN (2.56). This is surprising, given that TimeGAN scored few best S_A scores in the first instance. This points to a more stable performance across datasets for the TimeGAN model.

Table 2
Evaluation of model and representation by (a) number of best scores and (b) average ranks from 1 (best) to 5 (worst).

Rep.	# best S_D	# best S_P	# best S_A
TimeGAN	22	8	11
WGAN-GP (GASF)	19	**42**	11
WGAN-GP (XIRP)	17	41	17
WGAN-GP (Naive)	**25**	6	**45**
Diffusion	17	3	16
$\sum$	100	100	100

Model	# best S_D	# best S_P	# best S_A
TimeGAN	22	8	11
WGAN-GP	**61**	**89**	**73**
Diffusion	17	3	16
$\sum$	100	100	100

(a)

Mode/Rep.	$\bar{r}_{S_D}$	$\bar{r}_{S_P}$	$\bar{r}_{S_A}$
TimeGAN	2.96	3.49	2.56
WGAN-GP (GASF)	3.07	2.28	2.99
WGAN-GP (XIRP)	3.16	**2.05**	2.92
WGAN-GP (Naive)	3.34	3.47	**2.37**
Diffusion	**2.47**	3.7	3.63
TimeGAN	2.96	3.49	**2.56**
WGAN-GP	3.21	**2.6**	2.76
Diffusion	**2.47**	3.7	3.63

(b)

In the previous section, we identified that GASF and XIRP consistently perform well across diverse datasets—particularly those with stochastic or complex dynamics—highlighting their robustness in capturing both synthetic and real-world patterns. These trends are confirmed by the summary statistics in Table 2. GASF achieves the best S_P score in 45 out of 100 cases, followed closely by XIRP with 34. Naive and TS trail significantly, with Naive only achieving 10 best scores. The dominance of WGAN-GP is also clear: it accounts for 89 of the 100 best S_P scores, compared to just 11 for TimeGAN. When evaluating average ranks, GASF and XIRP again lead with values of 2.28 and 2.05, while TimeGAN, Naive, and Diffusion lag behind with ranks 3.4, 3.49, and 3.7, respectively. WGAN-GP maintains its lead with a better average rank (2.6) affirming the model's superiority in this metric.

In the previous section, we mentioned the robustness of the Naive representation and its solid performance on

Table 3
Ranking on distributional analysis across moments from 1 (best) to 5 (worst).

Model	TimeGAN	WGAN-GP (GASF)	WGAN-GP (XIRP)	WGAN-GP (Naive)	Diffusion
Air Quality	4.50	2.50	2.00	**1.50**	4.50
Bike Share	4.50	**1.75**	2.75	3.00	3.00
Brownian Motion	4.50	**1.75**	2.25	2.25	4.25
Energy Data	4.50	**2.00**	**2.00**	**2.00**	4.50
Merton Process	4.50	**2.00**	2.75	**2.00**	3.75
Noisy Sine	4.50	2.25	**2.00**	**2.00**	4.25
Power Law	3.75	3.25	**2.50**	**2.50**	3.00
Sine	3.50	2.50	**2.00**	3.00	4.00
Stock Data	4.00	2.75	**2.00**	2.75	3.50
Average rank	4.25	2.31	**2.25**	2.33	3.86

Rep.	Metric	σ	ADF	x_0	σ^2	ACF(1)	ACF(2)	ACF(4)	ACF(8)	JI
GASF	S_P	0.1587	0.1063	-0.0589	-0.1791	-0.0919	-0.1766	-0.2621	-0.3254	-0.0005
XIRP	S_P	0.0795	0.0979	-0.0630	-0.1779	-0.1348	-0.2290	-0.3322	-0.4237	0.0353
Naive	S_P	-0.1806	-0.0651	0.0168	0.1548	-0.1560	-0.1892	-0.2103	-0.2281	0.2171
GASF	S_D	-0.2876	-0.1939	-0.0472	0.2459	0.0239	0.0062	-0.0258	-0.0752	0.0409
XIRP	S_D	-0.2820	-0.1906	-0.0737	0.2154	0.0126	-0.0099	-0.0455	-0.0934	0.0412
Naive	S_D	-0.3780	0.1941	0.1520	-0.0150	0.5609	0.5350	0.4968	0.4291	-0.4645

Fig. 3. Correlation of S_D and S_P with time series characteristics across representations.

various datasets with respect to its augmentation quality. These results are confirmed by the ranking metrics: WGAN-GP and Naive representations consistently secure top average ranks for S_A, with Naive averaging just 2.37, ahead of GASF (2.99) and XIRP (2.92).

Regarding S_D, our ranking system reveals another interesting fact: In Table 1 the Diffusion model does not show any best score on S_D, but when we consult the average ranks in Table 2, we see that the Diffusion shows the most stable performance with an average rank of 2.47 across the datasets.

To further our analysis of distributional similarity, we now turn to Table 3 again for a detailed comparison. The table lists the average rankings of the considered models across nine datasets, based on their ability to replicate statistical moments (first to fourth) of the original time series distributions. The WGAN-GP (XIRP) model outperforms all others in terms of average rank, achieving the best overall score of 2.25, closely followed by WGAN-GP with GASF (2.31), and Naive (2.33). WGAN-GP consistently ranks on top across multiple datasets. In contrast, TimeGAN shows the weakest performance overall with an average rank of 4.25. The Diffusion model, while outperforming TimeGAN in average rank (3.86), falls short compared to all three WGAN-GP variants. Overall, the image-based WGAN-GP models demonstrate superior performance in preserving statistical moments of time series data, with XIRP and GASF encodings leading slightly ahead of the naive version. Table 6 in the Appendix also shows that all models are generating time series with meaningful statistics.

In the previous steps, we analyzed which representation is favorable with respect to S_D and S_P. Now, we investigate which time series characteristics influence the success of the representations in synthesis, contained in Fig. 3. To quantify the interdependence between the characteristics of the time series (variance, drift, etc.) and the metric scores, we calculate their correlation coefficient across datasets. In terms of the predictive score, both GASF and XIRP representations display positive correlations, albeit with varying strengths. GASF demonstrates moderate correlations with standard deviation, ADF, and ACF(4), while XIRP shows similar trends but with weaker correlations. Conversely, the naive representation exhibits a negative correlation with S_P, indicating weaker associations with the evaluated statistical properties. For the discriminative score, GASF and XIRP representations showcase negative correlations. GASF demonstrates moderate negative correlations with σ and x_0, whereas XIRP exhibits similar trends with slightly weaker associations. In contrast, the naive representation shows strong positive correlations with the test statistic, variance, autocorrelation lags 1 through 16, and jump intensity. Tables 1 and 2 show that the performance of WGAN-GP is at least competitive, and in many cases superior, to that of the TimeGAN model.

In addition to the choice of representation, the inversion method has emerged as a key factor influencing the success of the synthesis process. Visual inspections of synthetic time series obtained by IM and IRC show that IM returns smoother trajectories, which differ from the input time series and are thus easy to detect in the backtests. Fig. 4 compares the performance for each representation and inversion method for each S_D and S_P. A look at Fig. 4(a) shows that the IM inversion obtains a lower score on the discriminative metric. Since a high discriminative metric is best, we see that IRC improves the score for each representation. Improvements are most prominent in S_D, where both representations show gains from about 4.8%

in the naive case to around 360% for GASF and XIRP. For the predictive score, the improvements range from −3.9% for the naive representation to over 60% for XIRP and GASF. Again, we see that XIRP and GASF benefit almost to the same extent from IRC, while the naive model performs 3.9% worse.

The introduction of the IRC substantially improved performance regarding similarity and predictive ability, and the discovery of its superiority may also benefit other researchers in similar situations. Regarding computation times, we observed an advantage favoring WGAN and diffusion. On an Nvidia Tesla T4 GPU, training takes approximately 10 min for a single time series, including the backtests compared to around 25 min for the TimeGAN model. Note, however, that the initial implementation was not designed to run on a GPU and leaves room for improvement, so definitive statements on the training times are hard to make. The omnipresent tradeoff in machine learning between model performance and computation time also applies here; given more training epochs and increased network size, it is also likely that S_D can be improved such that the slight difference in the average ranks of S_D of around 5% flips in favor of the WGAN. Training hyperparameters such as batch size, epochs, and learning rates remained at the default values proposed in the TimeGAN paper throughout all experiments. Thus, we have a well-grounded statement on competitive performance. However, we cannot give all credit to the new representation, since comparing two inherently different neural networks always leaves uncertainty. Considering that the observed differences are substantial, we still conclude confidently that the proposed representation contributed to the improvement in performance. In appreciating this result, it is essential to recall that the WGAN-GP model has a substantially easier structure, with just one loss function and two networks, compared to five loss functions and five networks in the TimeGAN model.

An important aspect related to the computation times is the architectures of the WGAN-GP and the diffusion model. The diffusion model consists of a lightweight fully connected neural network (MLP) designed to predict added noise during the denoising process. It takes as input the concatenation of a noised feature vector and a normalized diffusion time step, and passes it through three layers: an input layer of size 29 (28 features + 1 time step), two hidden layers with 128 units each and ReLU activations, and an output layer of size 28 to match the feature dimensionality.

This WGAN-GP model is designed to generate synthetic 2D representations with a fixed sequence length of 28. The generator takes a noise vector of predefined dimension and transforms it through a dense layer, reshaping, and three upsampling blocks with convolutional layers, batch normalization, and LeakyReLU activations. The output is cropped to a final size of $28 \times 28 \times 1$. The discriminator receives $28 \times 28 \times 1$ images and processes them through four convolutional layers with increasing filter sizes (64 to 512), LeakyReLU activations, and optional dropout, followed by flattening and a final dense output. The model is trained using the Wasserstein loss with a gradient penalty, performing multiple discriminator steps per generator step. Adam optimizers with $\beta_1 = 0$ and $\beta_2 = 0.9$ are used for both networks. After training, synthetic samples are generated by feeding noise into the trained generator (see Fig. 3).

8. Future work

In future research, conducting a comprehensive analysis of the requisite data needed to adequately train each model and the benefits they offer given specific sample sizes is an interesting topic of investigation. Another approach would be to compare the results obtained with the WGAN-GP to other competitive approaches, such as diffusion models. A more thorough study of the statistical properties of the results could also answer the question of under which conditions a successful synthesis is possible. Calculating statistics (mean, variance, skewness, entropy, etc.) for all input datasets and relating them to models and representations could provide insights regarding which circumstances make it unlikely that augmenting the data will yield a benefit.

Further, it would be promising to follow the approach in the multivariate setting, similar to the one already used by TimeGAN, and to evaluate how well the models capture the relationships among the individual time series. As is common in computer vision, the representations could be easily stacked along a third axis and processed further. Increasing the dimensionality also motivates an extension of research on multivariate versions of the recurrence plots to determine which form is the most efficient encoding for this case. Another exciting path would be evaluating different data augmentation steps for the WGAN-GP. Since the WGAN-GP relies on image representation, many common preprocessing steps could improve the generative performance, such as cropping, zooming, or shifting. Choosing large filters to track long-range dependencies within a time series increases the representational capacity of the network, but doing so comes at the cost of a loss in computational and statistical efficiency. Since our research suggests that different representations best model different aspects of a time series, it would be conceivable to combine the strengths of the representations. Stacking multiple representations along a dimension is likely to give a potential model even more advantage compared to a classical time series model. However, this remains a subject of study and needs to be proven in further experiments as one of the many potential research paths and aspects originating from our findings.

9. Conclusion

The paper proposed a new two-dimensional image representation for time series—the extended intertemporal return plot (XIRP)—to leverage the advances of image-based generative models for time series. It preserves the dynamics in a scale-invariant way without losing the ability to invert a time series numerically to its original scale, a significant advantage compared to other image representations. The XIRP can thus function as a powerful representation in regression and classification tasks.

Relying on a Wasserstein GAN with gradient penalty (WGAN-GP) for the generation process, we validated the

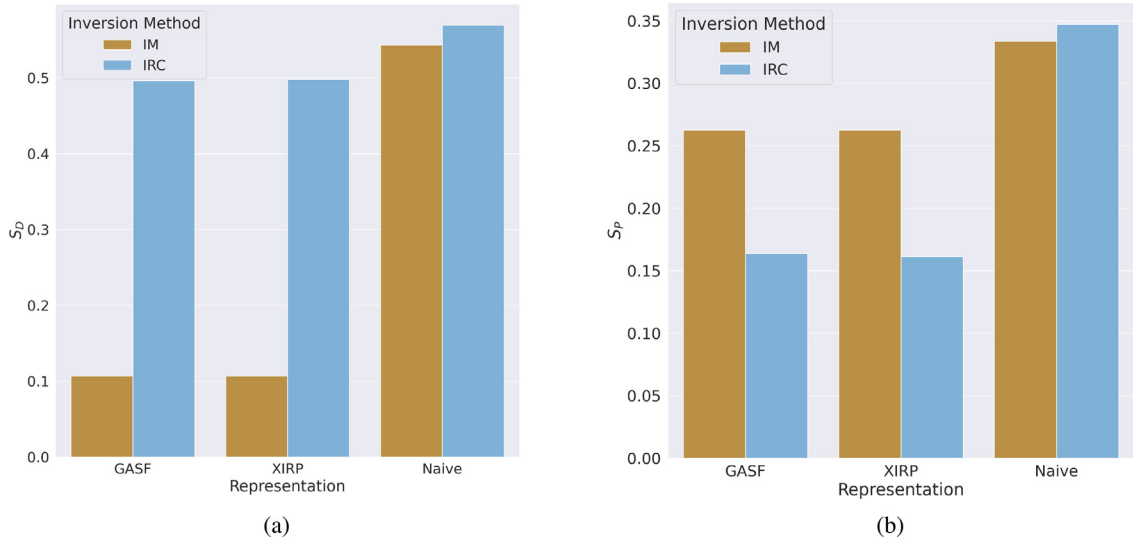


Fig. 4. Metric scores by representation and inversion method (IM: inversion by mean, IRC: inversion by random column) for (a) S_D (higher is better) and (b) S_P (lower is better).

XIRP on 100 time series from nine datasets regarding similarity and predictive ability measures. The results show that an off-the-shelf generative model trained on image-based representations successfully generates synthetic time series while reducing model complexity and computation time. Backed by our experiments, the newly proposed XIRP captures the dynamics and characteristics of the time series.

In terms of predictive ability, the proposed approach outperformed other image representations like the Gramian angular summation field (GASF) in a ranked performance comparison. Regarding forecast enhancement, the XIRP achieved a mediocre score outperformed by the naive representation and TimeGAN. Surprisingly, the WGAN-GP trained on a naive representation of column-wise stacked time series performed best on the similarity test and outperformed GASF, XIRP, and TimeGAN. In a cross-model comparison with an RNN-based TimeGAN and a diffusion model, the WGAN-GP trained on XIRPs outperformed them in terms of predictive ability and similarity of moments.

We further demonstrated that simulated data predominantly improved forecasting results, namely in four out of five models and on eight out of nine datasets. The set of tests conducted further indicated the absence of a universally superior representation across models, datasets, and evaluation metrics. This supports our broader conclusion that the choice of generative model for synthetic data should be guided by the specific purpose of generation.

Our findings also confirm that image-based generation is a viable alternative to RNN-based and diffusion-based models and can serve to complement their abilities regarding augmentation and similarity. Following our proposed ranking, we can state that invertible image representations such as XIRPs can capture temporal dynamics effectively and enable simple, off-the-shelf generative models to produce high-quality synthetic time series. Across all tests, synthesizing time series with an image-based model generally proved to be superior, outperforming the benchmarks in terms of predictive ability, distributional similarity, and forecast enhancement.

CRediT authorship contribution statement

Justin Hellermann: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Conceptualization. **Stefan Lessmann:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

Stefan Lessmann acknowledges financial support through the project "AI4EFin AI for Energy Finance", contract number CF162/15.11.2022, financed under Romania's National Recovery and Resilience Plan, Apel nr. PNRR-III-C9-2022-I8.

Appendix

See Tables 4–6 and Figs. 5–7.

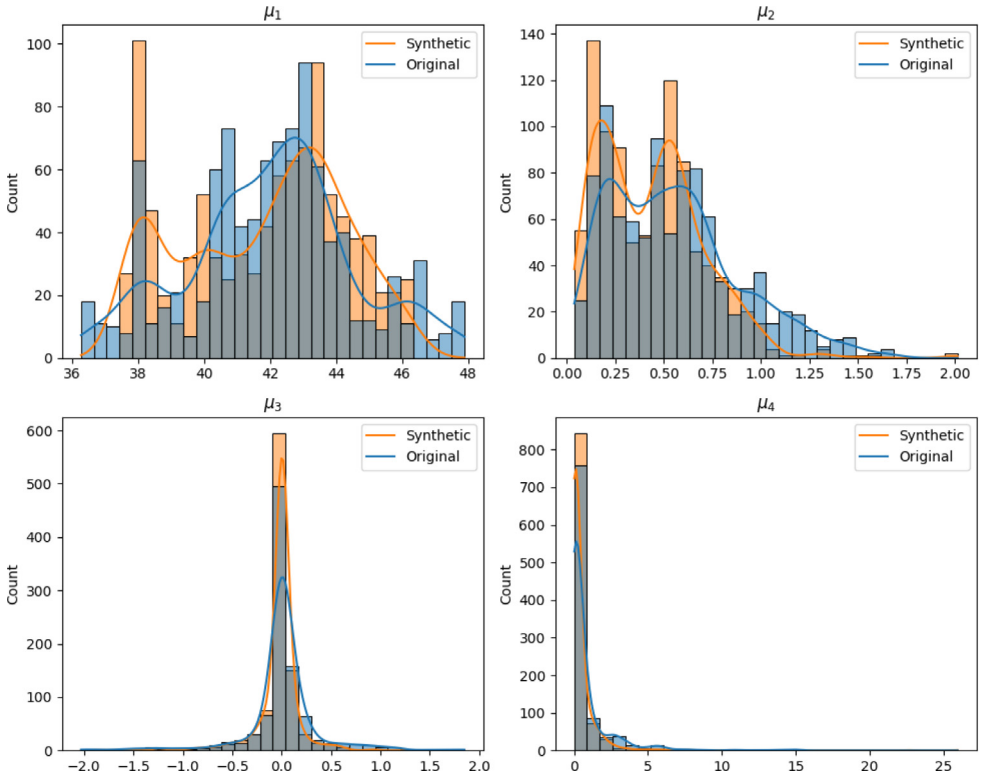


Fig. 5. The TimeGAN on Energy Data variable RH4 successfully synthesizes the data. In the analysis of the first moment, we observe a slight oversampling in the lower spectrum of the mean, accompanied by a subtle shift towards higher means. Similarly, the standard deviations exhibit comparable distributions, each displaying two peaks, with the original distribution showing a more pronounced tail.

Table 4
Image representations and characteristics.

	Invertible	Scale-invariant	Encoding	Modeling aspect	Classification	Regression	Feature scale
Binary recurrence plot	No	No	Deterministic	Exceedances	Yes	No	$x \in \mathbb{R}_{>0}$
Unthresholded recurrence plot	Yes, given x_0	No	Deterministic	Difference in raw values	Yes	Yes	$x \in \mathbb{R}_{>0}$
Markov transition fields	No	No	Probabilistic	Transition probabilities	Yes	No	$x \in \mathbb{R}$
Gramian angular summation field	Yes	No	Deterministic	Sum of polar coordinates	Yes	Yes	$0 \leq x \leq 1$
Gramian angular difference field	Yes, given x_0	No	Deterministic	Difference of polar coordinates	Yes	Yes	$-1 \leq x \leq 1$
Spectrograms	No	No	Probabilistic	Frequencies of occurrence	Yes	No	$x \in \mathbb{R}$
Waveform	No (approximation is possible)	No	Deterministic	Shape of its graph as a function of time	Yes	Yes	$x \in \mathbb{R}$
IRP	Yes, given x_0	Yes	Deterministic	Logarithmic percentage change	Yes	Yes	$x \in \mathbb{R}_{>0}$
XIRP	Yes, independent of x_0	Partially	Deterministic	Logarithmic percentage change and raw TS	Yes	Yes	$x \in \mathbb{R}_{>0}$

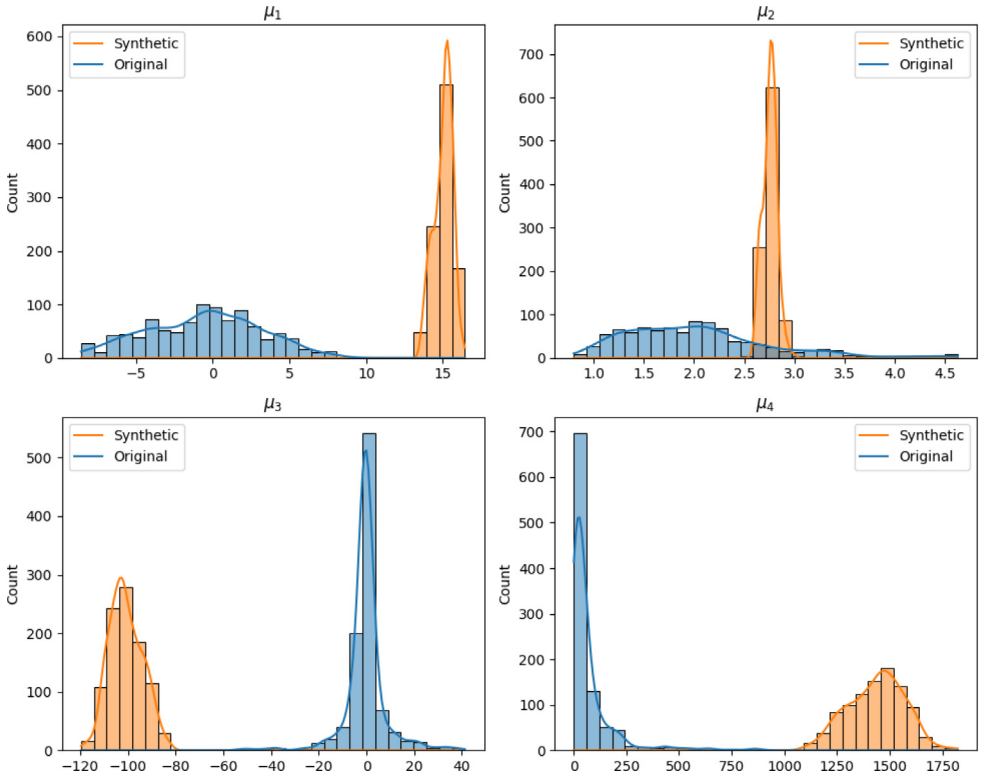


Fig. 6. The WGAN-GP with GASF on Noisy Sine with noise level 0.8 fails to reproduce the data accurately. Here, all four moments exhibit significant deviations. The discrepancy in the first moment can be attributed to a shift in the generated data. At the same time, the disparity in standard deviation suggests a much higher variance in the simulated data compared to the observed data. Furthermore, there is a notable difference in the distributions for all four moments. Possible explanations include premature termination of training due to an early stopping mechanism, or inadequate initialization for the given parameters. Multiple simulations for the same model configuration can help address flawed synthesis.

Table 5
Score by metric, representation, and inversion type.

Rep.	S	*IM*	*IRC*	$\Delta\%$
GASF	S_D	0.107	0.496	365.423
GASF	S_P	0.263	0.164	60.308
XIRP	S_D	0.107	0.498	366.619
XIRP	S_P	0.263	0.161	62.841
Naive	S_D	0.543	0.569	4.808
Naive	S_P	0.334	0.347	−3.881

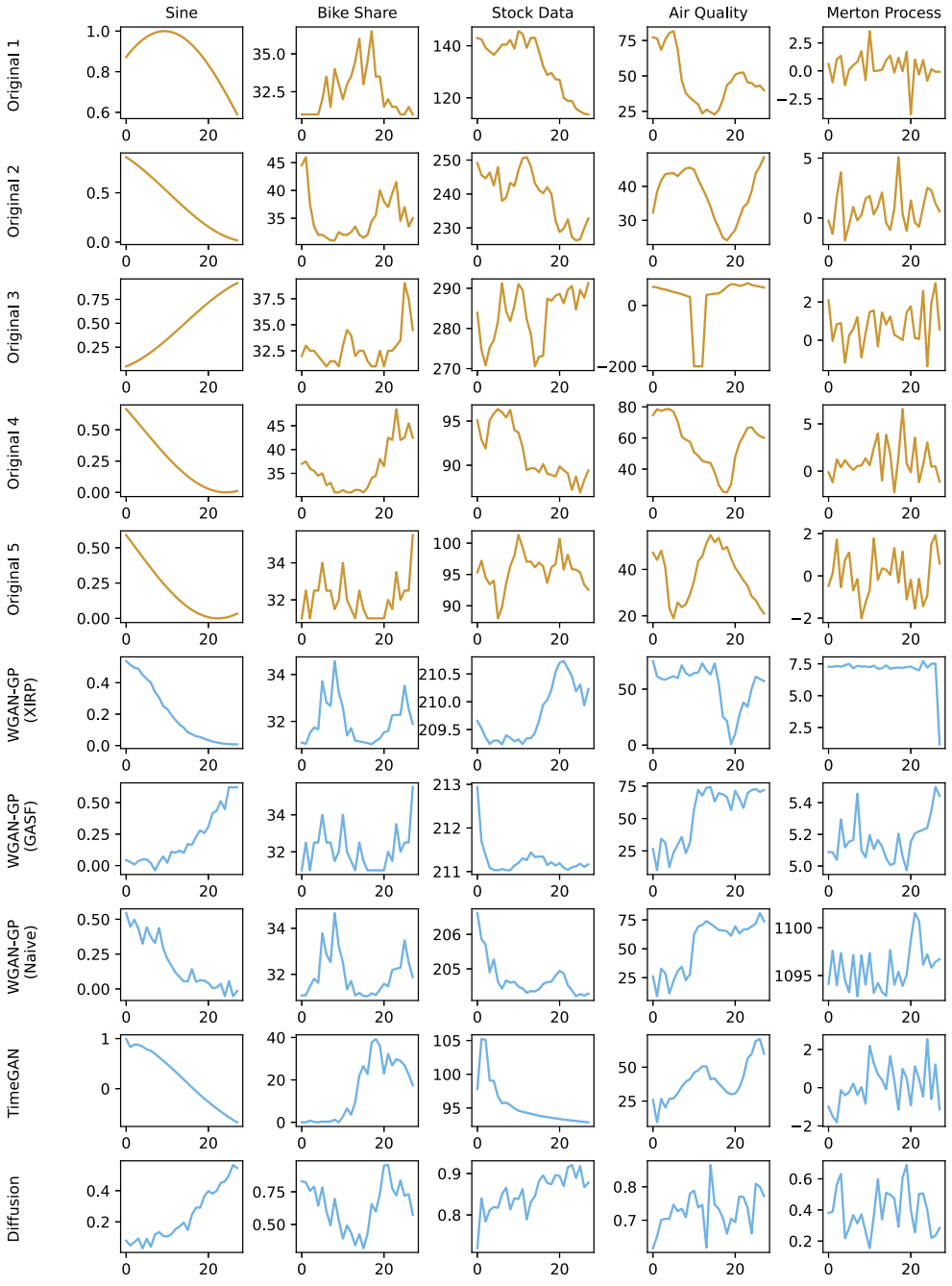


Fig. 7. A grid-based comparison between real and synthetic time series for five distinct data types: Sine, Bike Share, Stock Data, Air Quality, and Merton Process. The top five rows ('Original 1' to 'Original 5') show different real samples from the respective datasets. The subsequent rows visualize synthetic samples generated by the WGAN-GP (XIRP), WGAN-GP (GASF), WGAN-GP (Naive), TimeGAN, and Diffusion models. For each model and data type, the best-matching synthetic series is selected based on the minimum Wasserstein distance to a reference, here 'Original 5'. This approach allows for a direct visual assessment of how well each generative model captures the structure and dynamics of real-world time series data.

Table 6
Distribution of moments by dataset, model, and representation.

Dataset	Model	Rep.	$\tilde{\mu}_1(\mu)$	$\tilde{\mu}_1(\sigma)$	$\tilde{\mu}_2(\mu)$	$\tilde{\mu}_2(\sigma)$	$\tilde{\mu}_3(\mu)$	$\tilde{\mu}_3(\sigma)$	$\tilde{\mu}_4(\mu)$	$\tilde{\mu}_4(\sigma)$
Air Quality	Original	TS	7.1365e+02	7.5040e+01	8.6742e+01	3.2174e+01	1.0923e+06	2.4212e+06	2.1489e+09	2.7559e+09
	Diffusion	TS	7.5637e+02	7.6211e+01	8.8095e+01	3.2361e+01	1.1869e+06	2.5667e+06	2.3011e+09	2.7616e+09
	TimeGAN	TS	4.7642e+02	1.3712e+02	1.1334e+02	4.5162e+01	2.1500e+06	4.9200e+06	5.6653e+09	6.0322e+09
	WGAN-GP	GASF	7.7300e+02	5.9495e+01	7.8710e+01	2.9059e+01	7.7984e+05	1.7268e+06	1.1855e+09	1.8337e+09
		Naive	7.7292e+02	5.9732e+01	8.3490e+01	2.8486e+01	9.1809e+05	1.9155e+06	1.4572e+09	2.1100e+09
		XIRP	7.7293e+02	5.9386e+01	7.7394e+01	2.9333e+01	7.5363e+05	1.7317e+06	1.1339e+09	1.8355e+09
Bike Share	Original	TS	8.6377e+01	7.9339e+00	1.6733e+01	5.7438e+00	2.2552e+04	2.3271e+04	3.3292e+06	3.5802e+06
	Diffusion	TS	8.7948e+01	8.0794e+00	1.7242e+01	6.0452e+00	2.3526e+04	2.3949e+04	3.5329e+06	3.6301e+06
	TimeGAN	TS	2.5918e+01	2.0987e+01	1.8855e+01	6.8888e+00	3.6322e+04	5.4689e+04	9.2670e+06	1.1223e+07
	WGAN-GP	GASF	1.0150e+02	4.6438e+00	1.5987e+01	5.4945e+00	1.8843e+04	1.4866e+04	1.7780e+06	1.5652e+06
		Naive	1.0147e+02	4.7017e+00	1.6537e+01	5.3668e+00	1.9440e+04	1.6082e+04	1.9311e+06	1.7997e+06
		XIRP	1.0150e+02	4.6505e+00	1.5952e+01	5.5504e+00	1.8825e+04	1.4890e+04	1.7731e+06	1.5637e+06
Brownian M.	Original	TS	1.4052e+02	6.1262e+00	4.5210e+00	2.0561e+00	−1.0151e+04	4.4625e+04	8.1666e+06	3.9980e+07
.	Diffusion	TS	1.4179e+02	6.5050e+00	4.6939e+00	2.2583e+00	−1.0625e+04	4.8462e+04	8.7222e+06	4.1781e+07
	TimeGAN	TS	2.3973e−01	2.9214e+00	1.4910e+00	5.9598e−01	−3.1736e−01	5.3823e+00	3.4666e+01	7.3798e+01
	WGAN-GP	GASF	−3.7748e−01	2.0188e+00	1.8246e+00	6.7117e−01	−5.0484e−01	7.3236e+00	4.6794e+01	7.8659e+01
		Naive	1.5592e+03	3.0651e+00	2.3653e+00	7.0504e−01	−1.5058e+00	9.7849e+00	1.0540e+02	1.1992e+02
.		XIRP	5.6025e+00	1.2095e+00	1.8830e+00	5.2920e−01	−2.5860e+01	4.4541e+01	6.0912e+02	1.6908e+03
Energy Data	Original	TS	7.9476e+02	5.6616e+01	3.6174e+01	2.7439e+01	2.8393e+08	1.0473e+09	7.7825e+12	3.4371e+13
	Diffusion	TS	1.9493e+03	3.4944e+00	2.5850e+00	1.1852e+00	4.9512e+01	1.8196e+02	9.8761e+03	8.2958e+03
	TimeGAN	TS	5.6952e+01	3.4463e+00	1.6452e+00	6.1042e−01	−3.7510e+00	5.7173e+01	5.6546e+03	1.7138e+03
	WGAN-GP	GASF	6.6231e+01	3.9138e−01	2.4909e−01	1.1633e−01	−1.2519e−02	3.0496e−01	1.4240e+00	1.3284e+00
		Naive	1.8770e+03	3.4888e+00	2.5267e+00	1.1573e+00	4.6345e+01	1.7150e+02	9.1165e+03	8.1544e+03
		XIRP	7.0316e+01	6.8103e−01	1.3167e+00	2.1185e−01	−1.1506e+02	4.1458e+01	4.3628e+03	1.9111e+03
Merton P.	Original	TS	2.7341e+02	7.5998e+00	7.7678e+00	2.4094e+00	−1.0642e+04	1.9374e+04	4.6776e+06	8.4563e+06
	Diffusion	TS	5.4954e+00	1.9120e−01	4.4079e−01	3.1104e−02	−1.9783e+00	2.1992e−01	1.1374e+01	1.5974e+00
	TimeGAN	TS	4.6999e−01	1.2952e−01	9.1891e−01	1.3113e−01	1.9226e−01	4.9107e−01	4.5346e+00	2.8354e+00
	WGAN-GP	GASF	3.5828e+00	5.0575e−02	1.0032e−01	2.4199e−02	6.2918e−04	1.8632e−03	1.7475e−03	1.6222e−03
		Naive	6.7843e+02	1.0332e+00	1.0479e+00	1.8500e−01	5.7107e−01	9.5360e−01	1.0001e+01	7.8869e+00
		XIRP	5.2237e+00	1.8129e−01	4.3279e−01	2.8355e−02	−1.8360e+00	2.0103e−01	1.0440e+01	1.5073e+00
Noisy Sine	Original	TS	1.6439e+02	6.4008e+00	4.7341e+00	2.2135e+00	−2.4314e+03	6.7729e+03	8.2192e+05	2.7540e+06
	Diffusion	TS	4.4957e+02	9.7252e−01	7.2618e−01	1.5630e−01	−4.5489e−03	2.1639e−01	1.3518e+00	9.8433e−01
	TimeGAN	TS	1.2357e−02	6.5372e−01	5.1263e−01	1.0795e−01	−4.8808e−03	1.0633e−01	5.1970e−01	3.2010e−01
	WGAN-GP	GASF	1.7780e+00	1.1895e−01	7.6696e−02	3.3630e−02	−9.2614e−05	9.8299e−04	4.7380e−04	5.0073e−04
		Naive	4.1456e+02	9.0490e−01	6.8900e−01	1.4764e−01	−4.1485e−03	2.0401e−01	1.3155e+00	9.5601e−01
		XIRP	2.6696e+00	1.1705e−01	2.6529e−01	2.1436e−02	−3.4746e−01	4.1567e−02	1.0986e+00	1.7451e−01
Power Law	Original	TS	6.3387e+03	1.0602e+02	6.4150e+01	5.2634e+01	−8.5068e+06	7.2808e+07	1.3014e+11	6.7599e+11
	Diffusion	TS	1.6946e+02	8.8531e+01	3.7642e+00	2.1122e+00	1.3414e+02	6.9631e+02	1.3322e+04	3.2432e+04
	TimeGAN	TS	1.5731e+02	8.6806e+01	3.7621e+00	1.9529e+00	1.2528e+02	6.4900e+02	1.2368e+04	3.2194e+04
	WGAN-GP	GASF	1.4603e+02	7.0496e+00	1.4200e+00	1.0140e+00	−4.2880e+01	5.7387e+01	9.4782e+02	1.3084e+03
		Naive	1.5668e+04	1.1027e+01	3.4068e+01	5.3306e+00	−1.8895e+02	1.5031e+04	6.7498e+06	3.0797e+06
		XIRP	1.7996e+02	4.0282e+00	9.5985e+00	5.4830e−01	−2.1737e+04	2.0196e+03	2.7080e+06	3.3282e+05
Sine	Original	TS	8.1719e+01	1.0151e+01	5.7006e+00	3.8795e+00	−7.1385e+03	2.2532e+04	2.1812e+06	8.9227e+06
	Diffusion	TS	1.3650e+00	8.9562e−02	1.6509e−01	3.5926e−02	−3.1554e−02	1.0107e−02	4.0107e−02	1.7343e−02
	TimeGAN	TS	1.0362e−02	5.2905e−01	3.6307e−01	1.0205e−01	3.9301e−03	3.9851e−02	1.0777e−01	4.6865e−02
	WGAN-GP	GASF	8.5032e−01	7.7674e−02	3.6314e−02	1.9303e−02	−2.2955e−05	1.3711e−04	2.5415e−05	2.5609e−05
		Naive	1.9887e+02	7.6987e−01	4.8731e−01	1.7246e−01	4.0713e+03	6.2211e+02	1.9639e−01	1.7395e−01
		XIRP	1.2849e+00	8.6693e−02	1.6405e−01	3.4842e−02	−3.0560e−02	9.4194e−03	3.7703e−02	1.5930e−02
Stock Data	Original	TS	6.4694e+08	3.0647e+07	2.5822e+07	6.3471e+06	1.6257e+24	5.8250e+24	2.4055e+34	2.6868e+34
	Diffusion	TS	1.7613e+09	1.5893e+08	1.2773e+08	3.0932e+07	8.1615e+24	3.1194e+25	1.2160e+35	1.4024e+35
	TimeGAN	TS	3.2196e+06	1.6924e+06	9.5390e+05	5.2962e+05	1.7987e+19	3.5602e+19	9.2913e+26	1.4044e+27
	WGAN-GP	GASF	1.4707e+07	1.5046e+05	1.9187e+05	1.4699e+05	5.6574e+17	1.1480e+18	4.9954e+24	1.2591e+25
		Naive	1.5747e+09	2.7447e+06	2.1178e+06	8.0675e+05	3.9097e+20	4.9068e+20	2.0474e+28	3.4295e+28
		XIRP	1.8565e+07	4.2402e+05	1.0543e+06	9.2038e+04	−3.3912e+20	4.4886e+19	1.5809e+28	2.8516e+27

Data and code availability

The data that support the findings of this study are publicly available as described in the manuscript. The source code used for the analysis is openly accessible The data and code used in this study are openly available here.

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